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(54) **MARKING LABEL**

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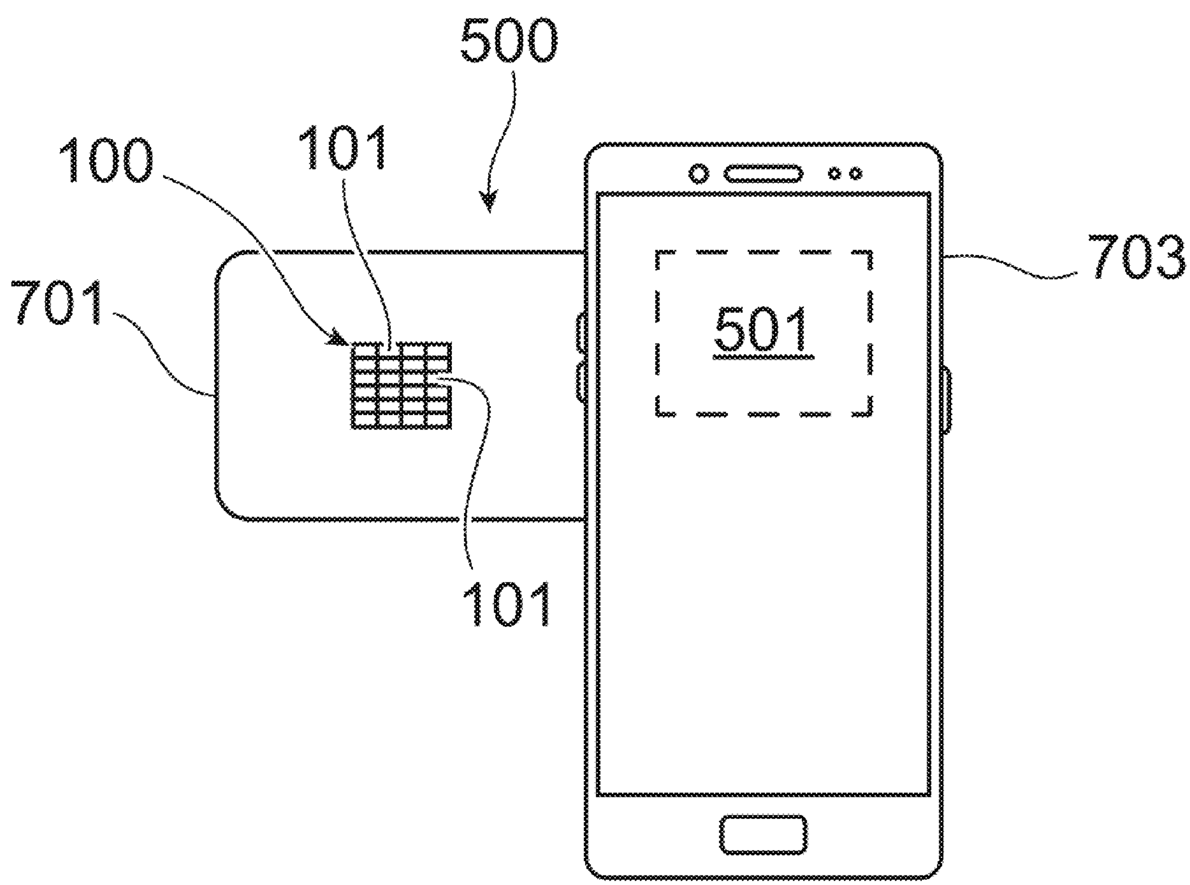
(57) **ABSTRACT**

An optical marking label includes one or more elementary cells. The elementary cells include first elementary cells which each are formed by at least one Fano resonance filter formed by a periodic structure made of a phase-change material. The elementary cells may include second elementary cells which are devoid of Fano resonance filters.



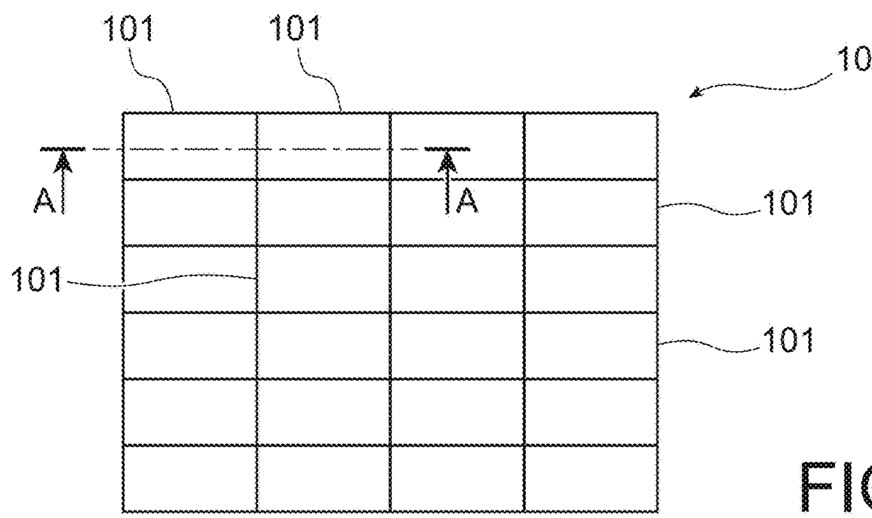


FIG. 1

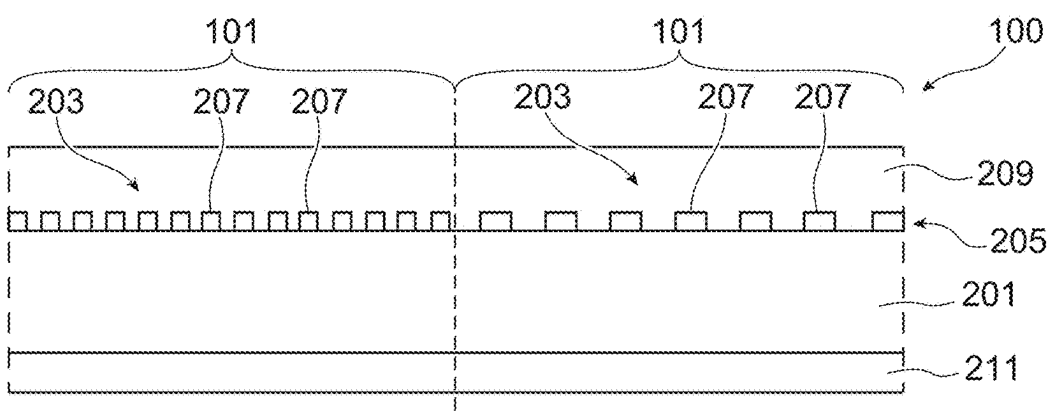


FIG. 2

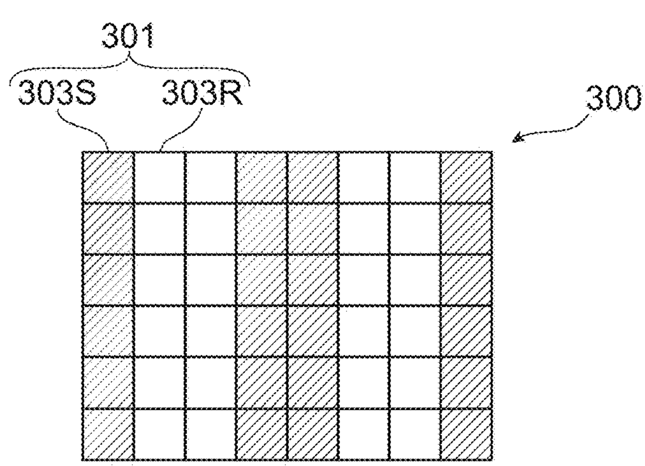


FIG. 3

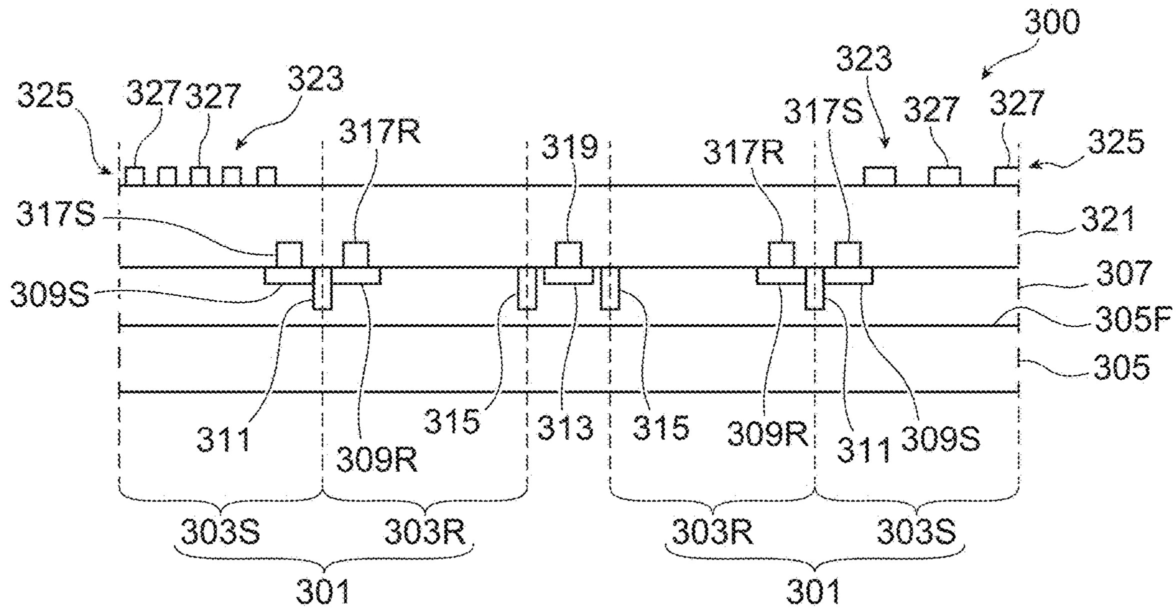

300
325
327
327
323
317R
319
317R
317S
323
327
327
325
317S
321
309S
307
305F
305
311
309R
315
313
315
309R
311
309S
303S
303R
303R
303S
301
301

FIG. 4

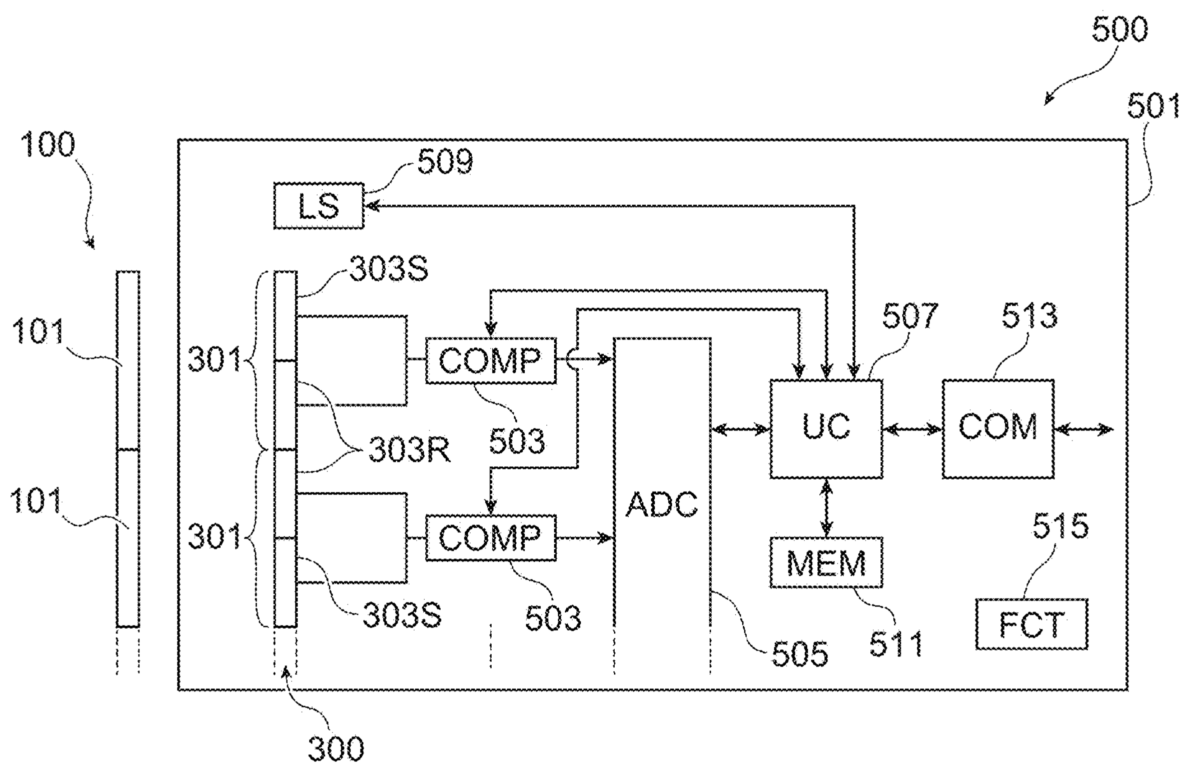

500
501
100
509
LS
303S
507
513
101
301
COMP
UC
COM
503
303R
101
ADC
515
301
COMP
MEM
303S
503
FCT
505
511
300

FIG. 5

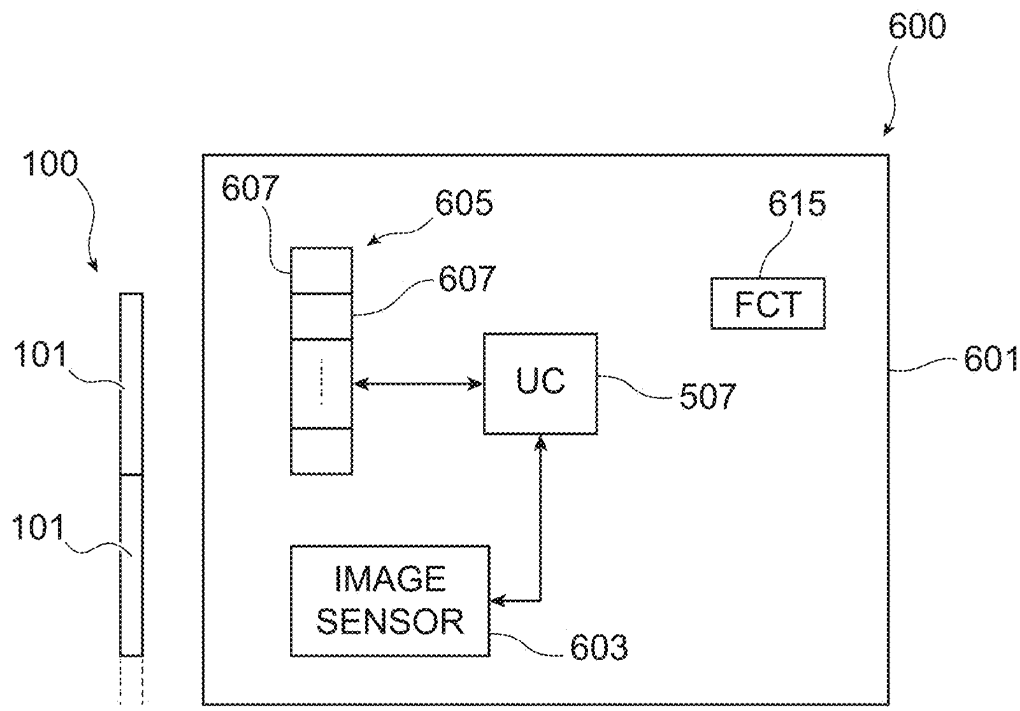

600
100
607
605
615
607
FCT
101
UC
601
507
101
IMAGE
SENSOR
603

FIG. 6

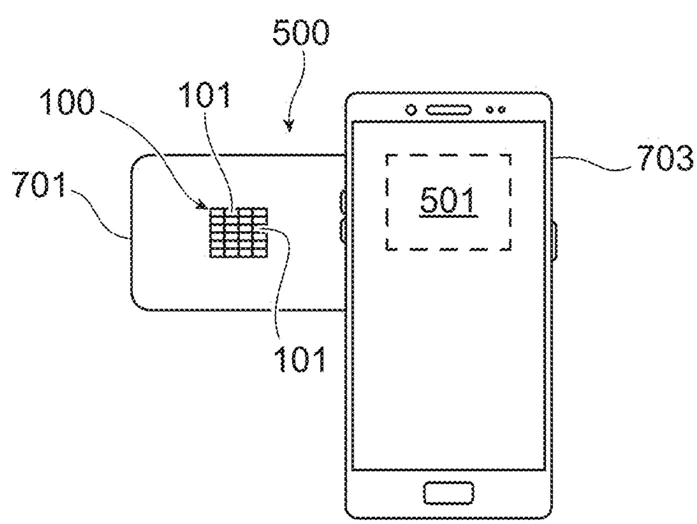

500
100
101
703
701
501
101

FIG. 7

MARKING LABEL

PRIORITY CLAIM

[0001] This application claims the priority benefit of French Application for Patent No. 2402206, filed on Mar. 5, 2024, the content of which is hereby incorporated by reference in its entirety to the maximum extent allowable by law.

TECHNICAL FIELD

[0002] The present disclosure generally concerns marking labels and, more particularly, marking labels which are passive, that is, devoid of electric power supply.

BACKGROUND

[0003] Many types of passive marking labels, typically enabling to identify and/or to authenticate products, identity documents, credit cards, etc., to which they have been affixed, have been provided. Among existing marking labels, passive labels comprising a one-dimensional (1D) barcode, for example an EAN (European Article Numbering) barcode, a two-dimensional (2D) barcode, for example a QR (Quick Response) code or a Datamatrix code, a security hologram, etc., have in particular been provided.

[0004] However, existing marking labels have various disadvantages. Existing marking labels are, in particular, likely to be falsified, compromising in this case the authenticity and/or the security of the marked products.

[0005] There exists a need to overcome all or part of the disadvantages of existing marking labels. It would in particular be desirable to have marking labels which cannot be falsified, or are very difficult to falsify, thus enabling to guarantee the authenticity and/or to increase the security of the marked products.

SUMMARY

[0006] In an embodiment, an optical marking label comprises one or a plurality of elementary cells, among which one or a plurality of first elementary cells each comprise at least one Fano resonance filter.

[0007] According to an embodiment, each Fano resonance filter comprises a periodic structure.

[0008] According to an embodiment, the periodic structure comprises an array of pads.

[0009] According to an embodiment, the periodic structure is made of a phase-change material.

[0010] According to an embodiment, said first elementary cell(s) each comprise a single Fano resonance filter.

[0011] According to an embodiment, the label comprises a plurality of first elementary cells with Fano resonance filters having different central frequencies.

[0012] According to an embodiment, said first elementary cell(s) each comprise a stack of at least two Fano resonance filters having different central frequencies.

[0013] According to an embodiment, said elementary cells further comprise one or a plurality of second elementary cells devoid of a Fano resonance filter.

[0014] An embodiment provides a device for reading the optical marking label such as described, the device comprising: a light sensor comprising one or a plurality of pixels arranged inside and on top of a semiconductor substrate, each pixel comprising a first photodetector comprising a Fano resonance filter, the central frequency or frequencies of the Fano resonance filter(s) of the readout device being substantially equal to the central frequency or frequencies of the Fano resonance filter(s) of the label; and a source of white light configured to illuminate the label.

[0015] According to an embodiment, each pixel of the light sensor further comprises a second photodetector devoid of a Fano resonance filter.

[0016] According to an embodiment, the light sensor further comprises at least one anode region common to a plurality of pixels.

[0017] According to an embodiment, the Fano resonance filter of the first photodetector comprises a periodic structure comprising an array of pads.

[0018] An embodiment provides a method of reading, by means of the readout device such as described, the optical marking label such as described, the method comprising the following steps: a) illuminating the label by means of the source of white light; and b) detecting, by means of the light sensor, the presence of the Fano resonance filter(s) in the label by comparison, with a reference signal, of an output signal of the photodetector of each pixel.

[0019] An embodiment provides a device for reading the optical marking label such as described, the device comprising: a visible image sensor intended to capture an image of the label; and a light source intended to illuminate the label, the light source comprising one or a plurality of elementary sources, the central emission frequency or frequencies of the elementary source(s) being substantially equal to the central frequency or frequencies of the Fano resonance filter(s) of the label.

[0020] An embodiment provides a method of reading, by means of the readout device such as described, the optical marking label such as described, the method comprising the following steps: a) illuminating the label by successively activating the elementary sources; b) capturing, by means of the visible image sensor, for each phase of illumination by one of the elementary sources, a visible image of the label; and c) detecting, based on the visible image(s), the presence and/or the position of the Fano resonance filter(s) in the label.

[0021] An embodiment provides a system comprising the label such as described and the readout device such as described.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The foregoing features and advantages, as well as others, will be described in detail in the rest of the disclosure of specific embodiments given as an illustration and not limitation with reference to the accompanying drawings, in which:

[0023] FIG. **1** is a top view, simplified and partial, of an example of a marking label;

[0024] FIG. **2** is a side and cross-sectional view along plane AA of FIG. **1**, simplified and partial, of the marking label of FIG. **1**;

[0025] FIG. **3** is a top view, simplified and partial, of an example of a light sensor;

[0026] FIG. **4** is a side and cross-sectional view, simplified and partial, of the light sensor of FIG. **3**;

[0027] FIG. **5** schematically and partially illustrates, in the form of blocks, an example of a marking label reading system;

[0028] FIG. **6** schematically and partially illustrates, in the form of blocks, another example of a marking label reading system; and

[0029] FIG. **7** illustrates, schematically and partially, an example of implementation of a marking label reading system.

DETAILED DESCRIPTION

[0030] Like features have been designated by like references in the various figures. In particular, the structural and/or functional features that are common among the various embodiments may have the same references and may dispose identical structural, dimensional and material properties.

[0031] For clarity, only those steps and elements which are useful to the understanding of the described embodiments have been shown and are described in detail. In particular, the different applications of marking labels have not been detailed, the described embodiments being compatible with all or most applications implementing marking labels, subject to possible adaptations within the abilities of those skilled in the art based on the indications of the present disclosure.

[0032] Unless indicated otherwise, when reference is made to two elements connected together, this signifies a direct connection without any intermediate elements other than conductors, and when reference is made to two elements coupled together, this signifies that these two elements can be connected or they can be coupled via one or more other elements.

[0033] In the following description, where reference is made to absolute position qualifiers, such as "front", "back", "top", "bottom", "left", "right", etc., or relative position qualifiers, such as "top", "bottom", "upper", "lower", etc., or orientation qualifiers, such as "horizontal", "vertical", etc., reference is made unless otherwise specified to the orientation of the drawings.

[0034] Unless specified otherwise, the expressions "about", "approximately", "substantially", and "in the order of" signify plus or minus 10%, preferably of plus or minus 5%.

[0035] In the following description, the qualifiers "insulating" and "conductive" respectively signify, unless indicated otherwise, electrically insulating and electrically conductive.

[0036] The expression "transmittance of a layer" designates a ratio between an intensity of a radiation coming out of the layer and the intensity of the radiation entering the layer. In the following description, a layer is said to be opaque to a radiation when its transmittance is, for this radiation, lower than 40%, preferably lower than or equal to 25%, more preferably lower than or equal to 10%. Besides, a layer is said to be transparent to a radiation when its transmittance is, for this radiation, higher than or equal to 40%, preferably higher than or equal to 75%, more preferably higher than or equal to 90%. The above definition of opaque and transparent is not limited to the case of a layer, but more generally applies to any element likely to be exposed to a radiation, for example a substrate, a region, a stack of a plurality of layers, etc.

[0037] In the present description, the expression "Fano resonance filter" designates a filter implementing an optical phenomenon called Fano resonance, or Fano effect. In such a filter, an incident radiation, for example visible light, irradiating a dielectric periodic structure of the filter excites a surface-confined mode supported by the surface of the periodic structure. This confined, or localized, mode interferes with the radiation reflected by the surface of the periodic structure. When the localized mode and the reflected radiation have the same phase, constructive interference between the localized mode and the reflected radiation leads to the appearing of a reflected radiation peak, that is, a radiation reflection peak. Thus, there exists a wavelength at which the incident radiation irradiating the filter is reflected at more than 80% or even at more than 90%, and this wavelength determines the central frequency of the filter. A Fano resonance filter thus behaves like a notch filter, conversely, for example, to a plasmonic filter.

[0038] In the present disclosure, the term "different Fano resonance filters" is used to designate Fano resonance filters having different central frequencies.

[0039] Examples of Fano resonance filters are described in the article by Shuai, et al., entitled "Double-layer Fano resonance photonic crystal filters", published in 2013 in Optics Express vol. 21, and in the article by Shen. et al. entitled "Structural Colors from Fano Resonances", published in 2014 in ACS Photonics (both of which are incorporated herein by reference).

[0040] Embodiments herein provide a marking label taking advantage of the above-mentioned Fano resonance phenomenon. An optical marking label is provided comprising one or a plurality of elementary cells, wherein the or at least one of the elementary cells comprises a Fano resonance filter. A device for reading optical marking labels comprising at least one Fano resonance filter is further provided.

[0041] FIG. **1** is a top view, simplified and partial, of an example of a marking label **100**. In the shown example, label **100** is more precisely an optical marking label. Marking label **100** is, in particular, devoid of electric power supply.

[0042] In the shown example, marking label **100** has, in top view, a substantially rectangular shape. This example is, however, non-limiting, and marking label **100** may more generally have, in top view, any shape, for example a polygonal shape other than rectangular—for example square, triangular, hexagonal, etc.—or a rounded shape—for example oval, circular, etc.

[0043] As an example, marking label **100** has a surface area in the range from 0.1 to 10 mm^2, for example equal to approximately 1 mm^2.

[0044] Marking label **100** comprises a plurality of elementary cells **101**. In the example shown in FIG. **1**, elementary cells **101** are arranged in the form of an array along rows and columns. FIG. **1** illustrates an example in which marking label **100** comprises twenty-four elementary cells **101** distributed in six rows and four columns. However, this example is not limiting, and marking label **100** may as a variant comprise any integer number N, greater than or equal to one, of elementary cells **101**. Further, if marking label **100** comprises a plurality of elementary cells **101**, these cells may be distributed in any manner within marking label **100**, for example arranged in an array comprising any non-zero integer numbers of rows and columns.

[0045] In the shown example, the elementary cells **101** of marking label **100** have substantially equal lateral dimensions, to within manufacturing dispersions. Further, elementary cells **101** are, in this example, distributed with a constant pitch, that is, with a constant center-to-center distance between two adjacent elementary cells **101**, along the rows and along the columns of the array of elementary cells **101**.

[0046] Although FIG. **1** illustrates an example in which elementary cells **101** are adjacent, this example is not limiting and marking label **100** may, as a variant, comprise one or a plurality of spaces devoid of elementary cells **101** laterally interposed between a plurality of cells or groups of elementary cells **101**. As an example, marking label **100** may, in this case, have a ring shape in which the elementary cells **101** are located in a peripheral region surrounding a blank central region, that is, a region which is devoid of elementary cells **101**.

[0047] In the shown example, each elementary cell **101** of marking label **100** has, in top view, a substantially rectangular shape. This example is however not limiting, and each elementary cell **101** may more generally have, in top view, any shape, for example a polygonal shape other than rectangular—for example square, triangular, hexagonal, etc.—or a rounded shape-for example oval, circular, etc.

[0048] FIG. **2** is a side and cross-sectional view along plane AA of FIG. **1**, simplified and partial, of marking label **100**. FIG. **2** more specifically corresponds, for example, to a side and cross-sectional view, along a plane substantially parallel to the rows or to the columns of elementary cells **101** of marking label **100**, of two adjacent elementary cells **101**.

[0049] In the shown example, each elementary cell **101** comprises a support substrate **201**. Support substrate **201** is, for example, made of a material transparent to visible light, for example made of glass.

[0050] According to an embodiment, at least one of the elementary cells **101** of marking label **100** comprises a Fano resonance filter **203**. In the shown example, the two elementary cells **101** each comprise a Fano resonance filter **203**. Each Fano resonance filter **203** comprises, for example, a periodic structure located on top of and in contact with a surface of support substrate **201** (the upper surface of support substrate **201**, in the orientation of FIG. **2**). The periodic structures of Fano resonance filters **203** are, for example, formed in an insulating layer **205** previously deposited on the upper surface of support substrate **201**. Each periodic structure, for example, forms a metastructure, also called metasurface.

[0051] As an example, the insulating layer **205** in which the periodic structures of the Fano resonance filter **203** are formed is made of amorphous carbon, of amorphous silicon, of silicon nitride, of undoped polysilicon, or of silicon carbide.

[0052] As a variant, insulating layer **205** may be made of a phase-change material, that is, of a material capable of alternating, under the effect of a temperature variation, between a crystalline phase and an amorphous phase. The phase-change material is, for example, a "chalcogenide" material, that is, a material or an alloy comprising at least one chalcogen element, for example a material from the family of germanium telluride (GeTe), of antimony telluride (SbTe), of antimony sulfide (Sb_2S_3), of antimony selenide (Sb_2Se_3), or of germanium-antimony-telluride (GeSbTe, commonly known by the acronym "GST"). The fact that the periodic structures of the Fano resonance filters **203** of marking label **100** are made of a phase-change material enables, for example, to obtain, for a same Fano resonance filter, two different central frequencies according to whether the material of insulating layer **205** is configured in the crystalline or amorphous state.

[0053] In the shown example, each Fano resonance filter **203** comprises an array of pads **207**, for example of substantially cylindrical shape with a circular cross-section. This example is, however, not limiting, and pads **207** may more generally have any shape, for example a cylindrical shape with a cross-section other than circular—for example square, triangular, hexagonal, oval, etc.—or a conical, frustoconical, pyramidal shape, etc. In top view, pads **207** are arranged in an array, for example, with a constant pitch, along rows and columns.

[0054] In order not to overload the drawing, only a few pads **207** have been symbolized in FIG. **2** for each Fano resonance filter **203**. However, this example is not limiting, and each Fano resonance filter **203** may, of course, comprise a number of pads much greater than what has been shown in FIG. **2**, for example several tens, several hundreds, or several thousands of pads **207**.

[0055] Within a single Fano resonance filter **203**, pads **207** have, for example, lateral dimensions and a pitch, that is, a center-to-center distance between two adjacent pads **207**, substantially constant, to within manufacturing dispersions. The Fano resonance filters **203** of the two elementary cells **101** illustrated in FIG. **2** have different central frequencies, and thus different rejection bands. In the illustrated example, the pads **207** of the Fano resonance filter **203** of the left-hand elementary cell **101** have lateral dimensions and a pitch different from those of the pads **207** of the Fano resonance filter **203** of the right-hand elementary cell **101**. This example is however not limiting, and the difference in central frequencies between two Fano resonance filters **203** may, as a variant, be obtained by modifying only the lateral dimensions or only the pitch of pads **207**, to within manufacturing dispersions.

[0056] Each pad **207**, for example, has a maximum lateral dimension—for example, a diameter, in the case where pads **207** are cylindrical—in the range from 100 to 600 nm. Further, the pitch of the array of pads **207** is, for example, in the range from 200 to 700 nm.

[0057] An example of embodiment of marking label **100** in which the periodic structure of each Fano resonance filter **203** comprises pads has been detailed hereabove. This example is however not limiting.

[0058] As a variant, the periodic structures of the Fano resonance filters **203** may each comprise a plurality of concentric rings made of an insulating material, for example selected from the materials listed hereabove for insulating layer **205**. In this case, the width and/or the pitch of the rings, that is, the distance separating the median circles of two adjacent rings, enable to define the central frequency of filter **203**.

[0059] As a variant, the periodic structures of the Fano resonance filters **203** may each comprise a plurality of vias or of cavities, for example of substantially cylindrical shape with a circular cross-section, formed in insulating layer **205**. This example is, however, not limiting, and the vias may more generally have any shape, for example a cylindrical shape with a cross-section other than circular—for example square, triangular, hexagonal, oval, etc.—or a conical, truncated cone shape. In this variant, the vias are, for example, arranged in an array, with a constant pitch, in rows and columns, the lateral dimensions and/or the pitch of the holes, that is, the center-to-center distance between two adjacent holes, enabling to define the central frequency of filter **203**.

[0060] In the shown example, the periodic structure of the Fano resonance filters **203** is defined by openings, or cavities, running across the entire thickness of insulating layer

205. However, this example is not limiting. As a variant, the periodic structure of each Fano resonance filter 203 may be defined by openings which do not run through insulating layer 205, for example openings extending in insulating layer 205 from a surface of layer 205 intended to be irradiated with a radiation (the upper surface of insulating layer 205, opposite to support substrate 201, in the orientation of FIG. 2) and interrupted in the thickness of layer 205. In this variant, the openings penetrate into insulating layer 205, for example, across substantially half of its thickness. Similarly, in the variants described hereabove where the periodic structures of the Fano resonance filters 203 comprise concentric rings or vias, the openings separating the rings or the openings forming the vias may by through or not.

[0061] In the illustrated example, each Fano resonance filter further comprises another insulating layer 209 coating insulating layer 205. Insulating layer 209 more specifically coats the side surfaces and the upper surface of pads 207. Insulating layer 209 is, for example, located on top of and in contact with the side surfaces and the upper surface of pads 207 and fills, that is, integrally fills, all the free spaces laterally extending between pads 207. Insulating layer 209 is made of a material different from that of pads 207. As an example, insulating layer 209 is made of silicon oxide or of silicon nitride.

[0062] Marking label 100 is, for example, intended to be illuminated by a radiation, for example visible light, on the side of the surface of support substrate 201 supporting the Fano resonance filter(s) 203. Although this has not been detailed in the drawings, the marking label is, for example, intended to be mechanically affixed to a substrate. For this purpose, an adhesive layer (not shown) may, for example, be provided on the side of a surface of the support substrate 201 opposite to its surface supporting the Fano resonance filter (s) 203.

[0063] Each elementary cell 101 further comprises, for example, an anti-reflection layer 211 coating a surface of support substrate 201 opposite to the Fano resonance filter 203 (the lower surface of support substrate 201, in the orientation of FIG. 2). In the illustrated example, anti-reflection layer 211 is located on top of and in contact with the lower surface of support substrate 201. As an example, anti-reflection layer 211 extends laterally in continuous manner over the entire lower surface of support substrate 201. Anti-reflection layer 211 has the function of avoiding for the incident radiation irradiating marking label 100, for example the visible light illuminating label 100, to be reflected after having crossed the Fano resonance filter(s) 203. In the case where an adhesive layer is provided to affix marking label 100 to a substrate, the adhesive layer coats, for example, a surface of anti-reflection layer 211 opposite to support substrate 201 (the lower surface of anti-reflection layer 211, in the orientation of FIG. 2).

[0064] Although this has not been detailed in FIGS. 1 and 2, each of the other elementary cells 101 of marking label 100 may either comprise a Fano resonance filter 203, or be devoid of a Fano resonance filter. As an example, in elementary cell(s) 101 devoid of a Fano resonance filter, insulating layer 205 is continuous and extends laterally over the entire surface of elementary cell 101. As a variant, insulating layer 205 is omitted in elementary cell(s) 101 devoid of a Fano resonance filter.

[0065] FIG. 2 illustrates an example in which at least two of the elementary cells 101 of marking label 100 each comprise Fano resonance filter 203. However, this example is not limiting. More generally, at least one of the elementary cells 101 of marking label 100 comprises Fano resonance filter 203, while the other cells 101 of label 100 may be either provided with or devoid of Fano resonance filter 203.

[0066] Marking label 100 comprises, for example, a non-zero integer number N, for example greater than or equal to two, of elementary cells 101 (N=24 in the example illustrated in FIG. 1). In this example, at least one of cells 101 and at most the N cells 101 comprise Fano resonance filter 203.

[0067] The central frequency of each Fano resonance filter 203 of label 100 is, for example, selected from a non-zero integer number C of possible central frequencies. As an example, the number C of possible central frequencies for each of the Fano resonance filters 203 is equal to the number N of elementary cells 101 of label 100. This example is, however, not limiting, and the number C may, as a variant, be different from number N, for example greater than number N.

[0068] As an example, the Fano resonance filters 203 of the elementary cells 101 of marking label 100 which are provided therewith have different central frequencies. As a variant, a plurality of Fano resonance filters 203 of marking label 100 may have substantially equal central frequencies.

[0069] By choosing, for each of the N elementary cells 101 of marking label 100, to integrate or not the Fano resonance filter 203, and by varying the central frequency of this filter from one cell to another, it is possible to obtain different configurations for label 100.

[0070] As an example, if all the Fano resonance filters 203 have different central frequencies (C≥N), and without taking into account the position in marking label 100 of the elementary cells 101 comprising Fano resonance filter 203, label 100 has 2^N-1 different possible configurations. In the example where label 100 comprises twenty-four elementary cells 101 (N=24), there are approximately sixteen million different possible configurations for label 100.

[0071] More generally, the number of possible configurations for marking label 100 can be adjusted: by modifying the total number of elementary cells 101 that label 100 comprises; by allowing, or not, for a plurality of Fano resonance filters 203 of label 100 to have the same central frequency; and/or by taking into account, or not, the position of the cells 101 comprising Fano resonance filter 203 in label 100.

[0072] In particular, the fact of allowing for a plurality of Fano resonance filters 203 to have the same central frequency and/or the fact of taking into account the position, in marking label 100, of the elementary cells 101 provided with Fano resonance filter 203 enables to increase the number of different possible configurations for label 100. Those skilled in the art are capable, based on the indications of the present disclosure, of adjusting the above parameters according to the application, in particular according to the number of different possible configurations desired for marking label 100.

[0073] Although this has not been detailed in the drawings, each elementary cell 101 may, as a variant, comprise a stack comprising at least two Fano resonance filters, for example similar or identical to filter 203, having different central frequencies. In the case where the position of the elementary cells **101** comprising Fano resonance filter **203** in marking label **100** is not taken into account, this enables, for example, as compared with the shown example where elementary cells **101** comprise a single Fano resonance filter, to decrease the lateral dimensions of marking label **100** while keeping the same number of different possible configurations for label **100**. In the case where the position of the elementary cells **101** comprising Fano resonance filter **203** in marking label **100** is taken into account, this enables, for example, as compared with the example shown where the elementary cells **101** comprise a single Fano resonance filter, to increase the number of different possible configurations for label **100**.

[0074] FIG. **3** is a top view, simplified and partial, of an example of a light sensor **300** according to an embodiment.

[0075] In the shown example, light sensor **300** comprises a plurality of pixels **301**. In the example shown in FIG. **3**, pixels **301** are organized in the form of an array along rows and columns. FIG. **3** illustrates an example in which light sensor **300** comprises twenty-four pixels **301** arranged in six rows and four columns. This example is, however, not limiting, and light sensor **300** may as a variant comprise any integer number M, greater than or equal to one, of pixels **301**. Further, in the case where light sensor **300** comprises a plurality of pixels **301**, these pixels may be distributed in any way within light sensor **300**, for example arranged in an array comprising any non-zero integer numbers of rows and columns. As a variant, light sensor **300** may comprise a single pixel **301**, for example in a case where label **100** comprises a single elementary cell **101** comprising Fano resonance filter **203**.

[0076] The number M of pixels **301** is greater than or equal to the number C of different central frequencies possible for the Fano resonance filters **203** of marking label **100**. As an example, the number M of pixels **301** of light sensor **300** is equal to the number N of elementary cells **101** of marking label **100**. This example is, however, not limiting, and the number M of pixels **301** may, as a variant, be different from the number N of elementary cells **101** of marking label **100**, for example greater than number N.

[0077] In the shown example, the array of pixels **301** of light sensor **300** has, in top view, a substantially rectangular shape. This example is, however, not limiting, and pixel array **301** may more generally have, in top view, any shape, for example a polygonal shape other than rectangular—for example square, triangular, hexagonal, etc.—or a rounded shape—for example oval, circular, etc. The shape of the array of pixels **301** is, for example, substantially identical to that of marking label **100**.

[0078] Further, in the shown example, each pixel **301** of light sensor **300** has, in top view, a substantially rectangular shape. This example is however not limiting, and each pixel **301** may more generally have, in top view, any shape, for example a polygonal shape other than rectangular—for example square, triangular, hexagonal, etc.—or a rounded shape—for example oval, circular, etc. The pixels **301** of light sensor **300** may have lateral dimensions different from those of the elementary cells **101** of marking label **100**.

[0079] In the shown example, the pixels **301** of light sensor **300** have substantially equal lateral dimensions, to within manufacturing dispersions. Further, in this example, the pixels **301** are distributed with a constant pitch, that is, with a constant center-to-center distance between two adjacent pixels **301**, along the rows and along the columns of the array of pixels **301**.

[0080] Each pixel **301** of light sensor **300** for example comprises a pair of photodetectors **303**S and **303**R, for example photodiodes. As an example, photodetector **303**S is a detection photodetector and photodetector **303**R is a reference photodetector. Photodetectors **303**S and **303**R are for example adapted to capturing visible light. The structure and the role of each photodetector **303**R, **303**S are discussed in further detail hereafter.

[0081] FIG. **4** is a side and cross-sectional view, simplified and partial, of light sensor **300**. FIG. **3** more specifically corresponds, for example, to a side and cross-sectional view, along a plane substantially parallel to the pixel lines **301** of light sensor **300**, of two adjacent pixels **301**.

[0082] In the shown example, pixels **301** are arranged inside and on top of a semiconductor substrate **305**. Semiconductor substrate **305** is, for example, a wafer or a piece of wafer, for example made of silicon. Semiconductor substrate **305** is doped with a first conductivity type, for example type P. In the illustrated example, pixels **301** are more specifically arranged on the side of a front surface **305**F of semiconductor substrate **305** (the upper surface of substrate **305**, in the orientation of FIG. **4**). The front surface **305**F of semiconductor substrate **305** is, for example, intended to receive visible light.

[0083] In the illustrated example, light sensor **300** comprises a semiconductor layer **307** coating semiconductor substrate **305**. Semiconductor layer **307** extends, for example, laterally and continuously on top of and in contact with the entire front surface **305**F of semiconductor substrate **305**. Semiconductor layer **307** is, for example, made of the same material as semiconductor substrate **305**. Semiconductor layer **307** is, for example, doped with the same conductivity type as semiconductor substrate **305** (type P, in this example) and has, for example, a doping level lower than that of semiconductor substrate **305**. As an example, semiconductor layer **307** is formed by epitaxial growth from surface **305**F of semiconductor substrate **305**.

[0084] In the shown example, each photodetector **303**R, **303**S comprises a cathode region **309**R, **309**S formed in semiconductor layer **307**. Each cathode region **309**R, **309**S extends, for example, in semiconductor layer **307** from a surface of semiconductor layer **307** opposite to semiconductor substrate **305** (the upper surface of layer **307**, in the orientation of FIG. **4**) down to a depth smaller than the thickness of semiconductor layer **307**. Cathode regions **309**R and **309**S are, for example, doped with a conductivity type opposite to that of semiconductor layer **307** (type N, in this example).

[0085] In the shown example, an insulating region **311** is laterally interposed between the cathode regions **309**R and **309**S of each pixel **301**. In the shown example, insulating region **311** extends in semiconductor layer **307** from the upper surface of semiconductor layer **307** down to a depth greater than that of cathode regions **309**R and **309**S, and smaller than the thickness of semiconductor layer **307**. As an example, insulating region **311** is an insulating trench.

[0086] In the shown example, an anode region **313** is formed in semiconductor layer **307**. Anode region **313** extends, for example, in semiconductor layer **307** from the upper surface of semiconductor layer **307** down to a depth smaller than the thickness of the semiconductor layer **307**. In the example illustrated in FIG. **4**, anode region **313** is located substantially at the center of the two shown pixels **301**. Anode region **313** is, for example, doped with the same conductivity type as that of semiconductor layer **307** (type P, in this example). In the illustrated example, anode region **313** is common to the photodetectors **303**S, **303**R of the two pixels **301**.

[0087] FIG. **4** illustrates an example in which anode region **313** is shared by, or placed in common between, two adjacent pixels **301**. This example is however not limiting. As a variant, each pixel **301** may have an anode region **313** distinct from the anode regions of the other pixels **301** of light sensor **300**, or the same anode region **313** may be common to a number of pixels **301** different from two, for example common to a group of four pixels **301**.

[0088] In the illustrated example, anode region **313** is surrounded by an insulating region **315** having for example, in top view, a ring shape. In the shown example, insulating region **315** extends in semiconductor layer **307** from the upper surface of semiconductor layer **307** down to a depth greater than that of cathode regions **309**R and **309**S, and smaller than the thickness of semiconductor layer **307**. Insulating region **315** has, for example, a height substantially equal to that of insulating region **311**. In the illustrated example, insulating region **315** is laterally separated from anode region **313** by a portion of semiconductor layer **307**. As an example, insulating region **315** is an insulating trench.

[0089] In the shown example, light sensor **300** further comprises cathode electrodes **317**S and **317**R located on top of and in contact with cathode regions **309**S and **309**R, respectively. Similarly, an anode electrode **319** is located on top of and in contact with anode region **313**. As an example, cathode electrodes **317**S and **317**R and anode electrode **319** are made of a conductive material, for example a metal or a metal alloy.

[0090] In the illustrated example, light sensor **300** further comprises an insulating layer **321** coating the surface of semiconductor layer **307** opposite to semiconductor substrate **305** (the upper surface of layer **307**, in the orientation of FIG. **4**). In this example, insulating layer **321** is more specifically located on top of and in contact with the upper surface of layer **307** and with the side and upper surfaces of cathode electrodes **317**S, **317**R and anode electrodes **319**. As an example, insulating layer **321** is made of silicon oxide, of silicon nitride, or of tetraethyl orthosilicate (TEOS).

[0091] In the shown example, photodetectors **303**S each comprise a Fano resonance filter **323**. The Fano resonance filters of the photodetectors **303**S of light sensor **300** are, for example, similar or identical to the Fano resonance filters **203** of marking label **100**. In particular, each Fano resonance filter **323** comprises a periodic structure formed in an insulating layer **325**. As an example, the periodic structures of the Fano resonance filters **323** of light sensor **300** each comprise an array of pads **327** similar or identical to the pads **207** of the Fano resonance filters **203** of marking label **100**. As a variant, the periodic structures of the Fano resonance filters **323** of light sensor **300** may each comprise concentric rings or a network of vias as discussed in further detail hereafter in relation with FIG. **2** for Fano resonance filters **203**.

[0092] In the shown example, photodetectors **303**R are devoid of Fano resonance filter **323**. In photodetectors **303**R, insulating layer **325** is, for example, omitted, as in the example illustrated in FIG. **4**, or extends laterally and continuously over the entire surface of insulating layer **321** opposite to semiconductor substrate **305**.

[0093] FIGS. **3** and **4** illustrate an example in which photodetectors **303**R are placed side by side along a same line of pixels **301**. In this example, each line of pixels **301** is, for example, formed by repetition of an elementary pattern consisting of two photodetectors **303**S located on either side of two adjacent photodetectors **303**R. This enables, for example, as previously discussed in relation with FIG. **4**, to share the anode region **313** of photodetectors **303**R and **303**S forming part of the two pixels **301** of each pair. This example is however not limiting. As a variant, each line of pixels **301** may be formed by repetition of an elementary pattern comprising a photodetector **303**R and an adjacent photodetector **303**S. This corresponds, for example, to the case where the anode region of the photodetectors **303**R and **303**S of a same **301** pixel is distinct from the anode regions of the photodetectors **303**R and **303**S of the other pixels **301**.

[0094] Although this has not been detailed in FIG. **4**, light sensor **300** may further comprise other elements, for example an anti-reflection layer interposed between semiconductor substrate **305** and the Fano resonance filters **323**, an optical device configured so that light reaches the Fano resonance filters **323** under a normal incidence, optical isolation walls laterally separating photodetectors **303**R and **303**S, etc.

[0095] Each photodetector **303**S is configured to receive, in a photosensitive area located in semiconductor layer **307**, light to be analyzed having all the operating wavelengths of light sensor **300** except for those which are located in the rejection band of the Fano resonance filter **323** that it comprises. Further, each photodetector **303**R is configured to receive, in a photosensitive area located in semiconductor layer **307**, light to be analyzed having all the operating wavelengths of sensor **300**. For each of photodetectors **303**S and **303**R, a readout circuit (not illustrated in FIG. **4**) delivers, for example, an output signal representative of the quantity of light to be analyzed received by the photodetector. Thus, the output signal of the photodetectors **303**S comprising Fano resonance filter **323** is representative of the proportion of light to be analyzed having all the operating wavelengths of the sensor except for those which are located within the rejection band of Fano resonance filter **323**. Similarly, the output signal of the photodetectors **303**R devoid of a Fano resonance filter is representative of the total proportion of light to be analyzed having all the operating wavelengths of sensor **300**.

[0096] As an example, light sensor **300** comprises, for each central frequency of the Fano resonance filter(s) **203** likely to be present in marking label **100**, at least one photodetector **303**S having its Fano resonance filter **323** exhibiting a central frequency substantially equal, to within manufacturing dispersions, to that of the Fano resonance filter **203** of label **100**. In this case, image sensor **300** for example comprises a number P of different Fano resonance filters **323** greater than or equal to the number C of different Fano resonance filters **203** of marking label **100**, the central frequencies of the different Fano resonance filters **323** of light sensor **300** being, for example, respectively equal, to within manufacturing dispersions, to those of the Fano resonance filters **203** of marking label **100**. This thus enables light sensor **300** to be capable of detecting the presence of all the different Fano resonance filters **203** likely to be present in marking label **100**. As a variant, light sensor **300** may be adapted to detecting only some of the Fano resonance filters **203** among the different Fano resonance filters **203** likely to be present in marking label **100**.

[0097] The detection, by light sensor **300**, of the presence of the Fano resonance filters **203** in marking label **100** is discussed in further detail hereafter.

[0098] FIGS. **3** and **4** illustrate an example in which light sensor **300** comprises as many photodetectors **303**R as photodetectors **303**S. This example is however not limiting. As a variant, light sensor **300** may comprise a number of photodetectors **303**R which is smaller than the number of photodetectors **303**S. In this case, sensor **300** comprises, for example, two or four times fewer photodetectors **303**R than photodetectors **303**S, or even a single photodetector **303**R.

[0099] FIG. **5** schematically and partially illustrates, in the form of blocks, an example of a system **500** for reading marking labels, for example the marking label **100** of FIGS. **1** and **2**, according to an embodiment. In particular, FIG. **5** illustrates a case in which the reading system takes no account of the position, in marking label **100**, of the elementary cells **101** comprising the Fano resonance filter **203**.

[0100] In the shown example, reading system **500** comprises a readout device **501** comprising light sensor **300**. Light sensor **300** is, for example, intended to be positioned in front of marking label **100**.

[0101] In the illustrated example, readout device **501** comprises comparators **503** (COMP). In this example, each comparator **503** is coupled, preferably connected, to the pair of photodetectors **303**R, **303**S of a same pixel **301**. Each comparator **503** is, for example, configured to deliver a signal or information representative of the proportion of the light to be analyzed having wavelengths located within the rejection band of the Fano resonance filter **323** of the photodetector **303**S to which it is coupled, or connected, based on the output signals of the photodetectors **303**S and **303**R of pixel **301**. To achieve this, comparator **503** is configured, for example, to subtract the value of the output signal of photodetector **303**R, or reference signal, for example a current I_R, from the value of the output signal of photodetector **303**S, for example a current Is. The result of this subtraction (I_R-I_S) is then representative of the proportion, or quantity, of light to be analyzed having wavelengths within the rejection band of the Fano resonance filter **323** of the considered pixel **301**. This thus enables readout device **501** to detect, for each pixel **301** of light sensor **300**, whether at least one of the elementary cells **101** of marking label **100** comprises a Fano resonance filter **203** having its central frequency substantially equal to that of the Fano resonance filter **323** of the considered pixel **301**.

[0102] Although this has not been shown in FIG. **5**, a current amplifier may be provided between each photodetector **303**S or **303**R and the associated comparator **503**.

[0103] As an example, readout device **501** is configured to apply a first correction factor to the value of the output signal of photodetector **303**S and/or a second correction factor to the value of the output signal of photodetector **303**R before subtracting them. The correction factor(s) is (are) determined, for example, after a step of calibration of light sensor **300**, for example in order to take into account, outside of the rejection band of the Fano resonance filter **323** of photodetector **303**S, differences in light transmission and/or in order to take into account manufacturing dispersions, for example surface differences, between the photosensitive areas of photodetectors **303**S and **303**R. As an example, the correction factor(s) are stored in a memory, for example a non-volatile memory, of readout device **501**. The implementation of the calibration step is within the abilities of those skilled in the art based on the indications of the present disclosure.

[0104] Further, readout device **501** is configured, for example, to normalize the values of the output signals I_S and I_R of photodetectors **303**S and **303**R with respect to the value of the output signal IR of photodetector **303**R. The normalization step is carried out, for example, after having applied a correction factor to the value of the output signal I_S of photodetector **303**S and/or a correction factor to the value of the output signal I_R of photodetector **303**R, or may be followed by a step of application of a first correction factor to the normalized output value of photodetector **303**S and/or a second correction factor to the normalized output value of photodetector **303**R. The correction factors are determined, for example, as previously discussed.

[0105] As an example, noting λ_{min} and λ_{max} as the minimum and maximum operating wavelengths, respectively, of light sensor **300** and noting λS_{min} and λS_{max} as the lower and upper wavelengths, respectively, of the rejection band of Fano resonance filter **323**, for example the −3 dB rejection band of filter **323**, the photosensitive area of photodetector **303**S receives light to be analyzed having wavelengths in the range from λ_{min} to λS_{min} and from λS_{max} to λ_{max}, while the photosensitive area of photodetector **303**R receives light to be analyzed having wavelengths in the range from λ_{min} to λ_{max}. Thus, output signal I_S representative of the quantity of light to be analyzed received by the photosensitive area of photodetector **303**S for wavelengths in the range from $\lambda_{min}$to λS_{min} and from λS_{max} to λ_{max}, output signal I_R being representative of the light to be analyzed received by the photosensitive area of photodetector **303**R for wavelengths in the range from $\lambda_{min}$to λ_{max}. Light sensor **300** then delivers, for example, information representative of the proportion of the light to be analyzed received by the sensor **300** which has the wavelengths in the range from λS_{min} to λS_{max}, for example by subtracting the value of the output signal I_S of photodetector **303**S from that of the output signal I_R of photodetector **303**R.

[0106] As an example, when light sensor **300** comprises a plurality of photodetectors **303**S with Fano resonance filters **323** having different central frequencies, sensor **300** is used to obtain information representative of the spectral distribution of light between different wavelength ranges, each corresponding to a rejection band of a Fano resonance filter **323**. Sensor **300** is thus, for example, an ambient light sensor (ALS). By analyzing the information representative of the spectral distribution of light, light sensor **300** enables to detect the presence or the absence, in marking label **100**, of the various Fano resonance filters **203** that it is likely to contain.

[0107] As an example, for each pixel **301** of light sensor **300**, the result of the subtraction of the value of the output signal I_S of photodetector **303**S from that of the output signal I_R of photodetector **303**R is compared with a threshold. In the case where the result is greater than or equal to the threshold, it is considered, for example, that marking label **100** comprises at least one elementary cell **101** having its Fano resonance filter **203** exhibiting a central frequency substantially equal to that of the Fano resonance filter **323** of the photodetector **303**S of the considered pixel **301**. However, if the result is lower than the threshold, marking label

100 is considered to contain no elementary cell **101** having its Fano resonance filter **203** exhibiting a central frequency substantially equal to that of the Fano resonance filter **323** of the photodetector **303**S of the considered pixel **301**.

[0108] Thus, marking label **100** may, depending on the different Fano resonance filters **203** that it comprises, be used to encode information, and light sensor **300** may be used to decode this information. As an example, in the case where the position of the Fano resonance filters **203** in marking label **100** is not taken into account, the C different Fano resonance filters **203** may be respectively associated with different binary weights, the presence or the absence of each filter being, for example, respectively associated, arbitrarily, with binary values 1 and 0. As an example, in a case where the different Fano resonance filters **203** of marking label **100** respectively correspond to distinct wavelength ranges of the visible light spectrum, the largest wavelength ranges correspond, for example, to most significant bits, while the smallest wavelength ranges correspond, for example, to least significant bits.

[0109] More generally, those skilled in the art will be capable, based on the indications of the present disclosure, of encoding a data item in marking label **100** and of implementing light sensor **300** to decode a data item contained in a marking label of the type of label **100**.

[0110] In the shown example, readout device **501** further comprises an analog-to-digital converter **505** (ADC) having the outputs of comparators **503** connected thereto. Converter **505** is configured, for example, to deliver a digital detection signal indicating, for each pixel **301** of light sensor **300**, whether marking label **100** comprises a Fano resonance filter **203** having its central frequency substantially equal to that of the Fano resonance filter **323** of the photodetector **303**S of the considered pixel **301**.

[0111] In the illustrated example, analog-to-digital converter **505** is connected to a control circuit **507** (UC), for example a microcontroller of readout device **501**. Control circuit **507** is, for example, configured to control light sensor **300** and to receive the output signal of analog-to-digital converter **505**.

[0112] In the shown example, control circuit **507** is further connected to a light source **509** (LS). Light source **509** is, for example, configured to illuminate marking label **100**, for example with white light. As an example, light source **509** comprises at least one light-emitting diode, for example a first light-emitting diode predominantly emitting red light, a second light-emitting diode predominantly emitting green light, and a third light-emitting diode predominantly emitting blue light. In this case, the first, second, and third light-emitting diodes of light source **509** are, for example, simultaneously turned on during a phase of illumination of marking label **100**.

[0113] In the illustrated example, readout device **501** further comprises a memory circuit **511** (MEM) connected to control circuit **507**. Memory circuit **511** enables, for example, to store operating parameters of readout device **501**, for example the correction factors determined during the step of calibration of light sensor **300**.

[0114] In the shown example, readout device **501** further comprises a communication interface **513** (COM). Communication interface **513** enables, for example, readout device **501** to exchange data, information, or signals with other elements or systems, not detailed in FIG. **5**.

[0115] Although this has not been detailed, readout device **501** may further comprise other elements and/or circuits symbolized, in FIG. **5**, by a functional block **515** (FCT).

[0116] FIG. **6** schematically and partially illustrates, in the form of blocks, another example of a system **600** for reading marking labels, such as the marking label **100** of FIGS. **1** and **2**. In particular, the reading system **600** of FIG. **6** differs from the reading system **500** of FIG. **5** in that reading system **600** takes into account the position, in marking label **100**, of the elementary cells **101** comprising the Fano resonance filter **203**.

[0117] In the shown example, reading system **600** comprises a readout device **601** comprising an image sensor **603** (IMAGE SENSOR). Image sensor **603** is, for example, intended to be positioned in front of marking label **100**. Image sensor **603** is, for example, adapted to forming visible images of marking label **100**. As an example, image sensor **603** is a color image sensor. As a variant, image sensor **603** may be a monochrome sensor.

[0118] In the illustrated example, readout device **601** further comprises a light source **605**. In this example, light source **605** comprises, for each of the different Fano resonance filters **203** likely to be present in marking label **100**, at least one elementary source **607**, for example a light-emitting diode, having a central emission frequency substantially equal, to within manufacturing dispersions, to the central frequency of the Fano resonance filter **203** of label **100**. In this case, light source **605** comprises, for example, a number K of different elementary sources **607** greater than or equal to the number C of different Fano resonance filters **203** of marking label **100**, the central frequencies of the different elementary sources **607** of light source **605** being, for example, respectively equal, to within manufacturing dispersions, to those of the Fano resonance filters **203** of marking label **100**. This thus enables reading system **600** to be capable of detecting the presence of all the different Fano resonance filters **203** likely to be present in marking label **100**. As a variant, reading system **600** may be adapted to detecting only part of the Fano resonance filters **203** among the different Fano resonance filters **203** likely to be present in marking label **100**.

[0119] In the shown example, readout device **601** further comprises control circuit **507** (UC). Control circuit **507** is, in readout device **601**, connected to visible image sensor **603** and to light source **605**. Control circuit **507** is, for example, configured to successively activate the different elementary sources **607** of light source **605** and to capture, by means of image sensor **603**, a visible image of marking label **100** at each phase of illumination by one of the elementary sources **607**. During each illumination phase, marking label **100** is, for example, shielded from ambient light so that marking label **100** is illuminated only, or predominantly, by the light generated by the corresponding elementary source **607**.

[0120] As an example, for each visible image acquired by image sensor **603**, the presence of at least one light region surrounded by a dark region indicates the presence and the position, in marking label **100**, of at least one elementary cell **101** with a Fano resonance filter **203** having a central frequency substantially equal to the central emission frequency of the elementary source **607** used to acquire the image. Further, the absence of a light region in each visible image acquired by image sensor **603** indicates the absence, in marking label **100**, of one or a plurality of elementary cells **101** with a Fano resonance filter **203** having a central frequency substantially equal to the central emission frequency of the elementary source **607** used to acquire the image.

[0121] Similarly to reading system **500**, reading system **600** enables to encode a data item in marking label **100** and to use readout device **601** to decode this data item. Reading system **600** however has the additional advantage of allowing the taking into account of the position of the Fano resonance filter(s) **203** in marking label **100** to encode the data.

[0122] As a variant, readout device **601** may be used to detect the presence only, in marking label **100**, of at least one elementary cell **101** with a Fano resonance filter **203** having a central frequency substantially equal to the central emission frequency of the elementary source **607** used to acquire the image. In this variant, the position of the Fano resonance filter(s) in marking label **100** is not taken into account.

[0123] Although this has not been detailed in FIG. **6**, readout device **601** may further comprise other elements and/or circuits, for example a memory circuit similar or identical to the memory circuit **511** of device **501**, a communication interface similar or identical to the communication interface **513** of readout device **501**, etc. These elements and/or circuits have been symbolized in FIG. **6** by a functional block **615** (FCT).

[0124] FIG. **7** illustrates, schematically and partially, an example of implementation of a system for reading a marking label, for example the system **500** for reading marking label **100**, according to an embodiment.

[0125] In the shown example, marking label **100** is affixed to a card **701**, for example a bank card, an identity card, a personal identification card, a transport card, etc. This example is however not limiting, and marking label **100** may, as a variant, be affixed to any type of support, device, or product, for example a fashion item, such as a garment or a handbag, an electronic device, a passport, a banknote, a driving license, a card for accessing a secure area, a critical spare part, for example in the field of motor vehicles or of avionics, an ink cartridge for a printer, etc.

[0126] In the illustrated example, the readout device **501** of reading system **500** is integrated in a cell phone **703**. In this case, the light source **509** of readout device **501** corresponds, for example, to a light source of cell phone **703** used as a flash, in combination with a visible image sensor (not detailed) of the cell phone **703**, or as a torch lamp. Further, light sensor **300** corresponds in this case, for example, to an ambient light sensor of cell phone **703**.

[0127] FIG. **7** illustrates a case in which readout device **501** is integrated in a cell phone. This example is, however, not limiting, and readout device **501** may, as a variant, be integrated in other electronic devices such as a touch-sensitive tablet, a laptop, a device specifically dedicated to reading marking labels of the type of label **100**, etc. Further, although FIG. **7** illustrates an example of the implementation of reading system **500**, this example is not limiting and those skilled in the art are capable, based on the indications of the present disclosure, of integrating reading system **600** into various types of devices, for example selected from among the above-mentioned examples. Those skilled in the art are in particular capable of substituting the readout device **501** of reading system **500** with the readout device **601** of reading system **600** in cell phone **703**. In this case, the image sensor **603** of readout device **601** corresponds, for example, to a visible image sensor of cell phone **703**. The integration of readout device **501** into cell phone **703** is, for example, easier than the integration of readout device **601**, since advantage can be taken of the presence of components already existing in cell phone **703** to implement the functions implemented by readout device **501**.

[0128] The use of Fano resonance filters in the marking label and/or in the associated readout device enables to have marking labels and/or readout devices very difficult to falsify, since Fano resonance filters are formed by complex and expensive microelectronic equipment. This enables to guarantee the authenticity and/or to reinforce the security of marked products. It is indeed unlikely that an individual, or a small group of individuals, would manage to gather the technical means required to manufacture the marking labels and/or the readout devices of the present disclosure.

[0129] Various embodiments and variants have been described. Those skilled in the art will understand that certain features of these various embodiments and variants may be combined, and other variants will occur to those skilled in the art. In particular, although examples of reading systems **500** and **600** have been detailed in relation with FIGS. **5** and **6**, the embodiments are not limited to these examples and those skilled in the art are capable, based on the indications of the present disclosure, of providing reading systems different from those detailed, in particular readout devices different from those detailed.

[0130] Finally, the practical implementation of the described embodiments and variants is within the abilities of those skilled in the art based on the functional indications given hereabove. In particular, those skilled in the art are capable, based on the indications of the present disclosure, of adapting marking label **100** to the coding of different information or data, for example information linked to a product—such as a serial number, a brand name, or an item reference—to an authenticity control, etc.

[0131] Further, those skilled in the art are capable, based on the indications of the present disclosure, of defining the pitch and the lateral dimensions of the periodic structures of Fano resonance filters to obtain the desired central wavelengths. Further, the forming of light source **605**, in particular of elementary sources **607**, is within the abilities of those skilled in the art based on the indications of the present disclosure.

[0132] Those skilled in the art are also capable of adapting the operation of reading systems **500** and **600** to the case where the periodic structures of the Fano resonance filters **203** of marking label **100** are made of a phase-change material. In this case, two successive reading phases respectively corresponding to the two states—amorphous and crystalline—of the phase-change material may be implemented to detect the presence of the Fano resonance filters **203** in marking label **100**.

1. An optical marking label, comprising:

a plurality of elementary cells;

wherein one or more first elementary cells of the plurality of elementary cells each comprise at least one Fano resonance filter formed by a periodic structure made of a phase-change material.

2. The label according to claim **1**, wherein the periodic structure comprises an array of pads.

3. The label according to claim **1**, wherein said one or more first elementary cells each comprise a single Fano resonance filter.

4. The label according to claim **1**, wherein said one or more first elementary cells comprise a plurality of first elementary cells with Fano resonance filters having different central frequencies.

5. The label according to claim **1**, wherein said one or more first elementary cells each comprise a stack of at least two Fano resonance filters having different central frequencies.

6. The label according to claim **1**, wherein said plurality of elementary cells further comprise one or more second elementary cells, wherein each second elementary cell is devoid of a Fano resonance filter.

7. A device for reading an optical marking label comprising a plurality of elementary cells, wherein one or more first elementary cells of the plurality of elementary cells each comprise at least one Fano resonance filter formed by a periodic structure made of a phase-change material, the device comprising:

a light sensor comprising one or a plurality of pixels arranged inside and on top of a semiconductor substrate, each pixel comprising a first photodetector comprising a Fano resonance filter, wherein a central frequency or frequencies of the Fano resonance filter(s) of the light sensor are substantially equal to a central frequency or frequencies of the Fano resonance filter(s) of the optical marking label; and

a source of white light configured to illuminate the optical marking label.

8. The device according to claim **7**, wherein each pixel of the light sensor further comprises a second photodetector devoid of a Fano resonance filter.

9. The device according to claim **7**, wherein the light sensor further comprises at least one anode region common to a plurality of pixels.

10. The device according to claim **7**, wherein the Fano resonance filter of the first photodetector comprises a periodic structure comprising an array of pads.

11. A device for reading an optical marking label comprising a plurality of elementary cells, wherein one or more first elementary cells of the plurality of elementary cells each comprise at least one Fano resonance filter formed by a periodic structure made of a phase-change material, the device comprising:

a visible image sensor intended to capture an image of the optical marking label; and

a light source configured to illuminate the label, the light source comprising one or a plurality of elementary sources, wherein a central emission frequency or frequencies of the elementary source(s) are substantially equal to a central frequency or frequencies of the Fano resonance filter(s) of the optical marking label.

12. A system, comprising:

an optical marking label comprising a plurality of elementary cells, wherein one or more first elementary cells of the plurality of elementary cells each comprise at least one Fano resonance filter formed by a periodic structure made of a phase-change material; and

a device for reading an optical marking label, comprising:

a light sensor comprising one or a plurality of pixels arranged inside and on top of a semiconductor substrate, each pixel comprising a first photodetector comprising a Fano resonance filter, wherein a central frequency or frequencies of the Fano resonance filter(s) of the light sensor are substantially equal to a central frequency or frequencies of the Fano resonance filter(s) of the optical marking label; and

a source of white light configured to illuminate the optical marking label.

13. A method of reading an optical marking label comprising a plurality of elementary cells, wherein one or more first elementary cells of the plurality of elementary cells each comprise at least one Fano resonance filter formed by a periodic structure made of a phase-change material, the method comprising:

a) illuminating the optical marking label by means of a source of white light; and

b) detecting, by means of a light sensor, presence of the Fano resonance filter(s) in the optical marking label by comparing an output signal of the photodetector of each pixel with a reference signal.

14. A method of reading an optical marking label comprising a plurality of elementary cells, wherein one or more first elementary cells of the plurality of elementary cells each comprise at least one Fano resonance filter formed by a periodic structure made of a phase-change material, the method comprising:

a) illuminating the optical marking label by successively activating one or a plurality of elementary sources;

b) capturing, by means of a visible image sensor, for each phase of illumination by one of the one or a plurality of elementary sources, a visible image of the optical marking label; and

c) detecting, based on the visible image(s), a presence and/or position of the Fano resonance filter(s) in the optical marking label.

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